

 Solapur University, Solapur Sangola College, Sangola Dist. Solapur Solapur University Credit and Grading System Internal Examination/ Test/ Tutorial/ Home Assignment/ Practical Examination/ Preliminary Exam				
Name of Examination:			Sign. of Jr Supervisor	
Class:		Subject:		
Sem. No:		F. No:		
Roll No:		S. No:		
Day & Date:		Total Marks:		
Que. No.	Marks Alloted	Marks Obtained	Signature & Name of the Teacher	Signature of the Student
1				
2				
3				
4				
5				
Total:				

① Computer network is a group of computers connected with each other through wires, optical fibres or optical link so that various devices can interact with each other through a network.

② The aim of the computer network is the sharing of resource among various devices.

③ In the case of computer network technology, there are several types of network that vary from simple to complex level.

* Components of computer network.

① NIC (Network interface card)

NIC is a device that helps the computer to communicate with another device. The network interface card contain the hardware address, the data link layer use this address to identify the system on the network so that it transfer the data to the correct destination.

② Hub- Hub is a central device that splits the network connection into multiple devices when computer request for information from a comp. it sends the request to the hub. Hub distributes the request to all the interconnected computers.

Switches - Switch is networking device that groups all the devices over the network to transfer the data to another device. a switch is better than Hub as it does not broadcast the message over the network. i.e it sends the message to the device for which belongs to. Therefore, we can say that switch sends the message directly ~~to the~~ from source to the destination.

Router - Router is a device that connect LAN To the network. The router is mainly used to connect distinct network or connect the internet to multiple computers

Modem $\Rightarrow$ modem connects the computer to the internet over the existing telephone line. A modem is not integrated with computer motherboard. A modem is a ~~separate~~ separate part on the p.c. Slots found on the motherboard.

* Uses of computer Network $\Rightarrow$

$\rightarrow$ (i) Resource Sharing $\Rightarrow$

Resource Sharing is the sharing of resources such as program, printers and data among the users on the network without the requirement of the physical location of resource and as

(ii) Server client model - computer networking is used in the client server model.

a server is a central computer used to store the information & maintained by the system administrator clients are the machine used

Solapur University, Solapur

Sandola College, Sandola Dist Solapur
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used to access the information store in the server remotely.

* Computer network

Communication medium - Computer network behaves as a communication medium along users. ex. a company contains more than one computer has an email system which employees uses for daily communication.

* E-Commerce - Computer network is also important in business. we can do the business over the internet for example amazon.com is doing their business over the internet i.e. they are doing business on internet.

Types of Computer =>

① personal area network

PAN is the most basic type of computer network. It is a type of network designed to connect devices within short range typically around one person. It allows your personal devices, like smartphone, tablets, laptops and wearables to communicate and share data with each other. PAN offers a network range of 1 to 100 m from person to device providing communication. Its transmission speed is very high with very easy maintenance and very low cost. This uses Bluetooth. Examples of PAN are USB, computer, phone, tablet.

Advantages -> ① PAN is relatively flexible & provides high efficiency for short network range.

② It needs easy set up and relatively low cost.

③ It does not require frequent installation & maintenance.

④ It is easy and portable.

⑤ Disadvantages of PAN:-

① Low network coverage area

② Limited to relatively low data rates

Application ① Home & offices.

② Medical & Hospital

③ Organization & the Business.

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Solapur University Credit and Grading System

Internal Examination of Test/Tutorial/Work/Assignment/Practical
Examination/Preparatory Exam

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Class: _____ Page No: _____

Sem. No: _____ Seat No: _____

Roll No: _____ (Roll Name): _____

Day & Date: _____

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① LAN - LAN stands for local area network, which is a network of computers & other electronic devices that are all located in the same physical area. LAN can be found almost everywhere, including homes, public workplaces & public libraries, school college area etc. The purpose of LAN is to allow computers & other peripherals located in the same area to connect & exchange information with one another easily. In LAN, devices are linked to a network either be wired or wireless. In case of wired network we use ethernet cable & in wireless connection use WIFI.

Advantages of LAN -

① Privacy - LAN is private network thus no outside regulatory body controls it giving privacy.

② High Speed - LAN offers a much higher speed and data transfer rate comparatively to WAN.

③ Supports different transmission medium.

LAN support a variety of communication transmission medium such as Ethernet cable, fiber & wireless transmission.

~~disadvantages~~ Inexpensive & simple - A LAN usually has low cost, installation, expansion & maintenance & LAN installation is relatively easy to use.

Disadvantages of LAN →

- ① The initial setup cost of installing Local Area network is high. because there is special software required to make a server.
- ② communication devices like an ethernet cable switches, hubs router are costly.
- ③ LAN are restricted in size & cover only limited area.

* MAN → MAN is larger than LAN but smaller than WAN. This is the type of computer network that connects computer over a geographical distance through a shared communication path over a city, town. ex. ATM

The technology with range from 5km to > 50km. Its transmission speed is average. It is difficult to maintain & it comes with high cost. examples of MAN networking in towns, city.

- Adv - ① MAN offers high speed connectivity in which speed ranges from 10-100mbps
- ② The security level in MAN is high & strict as compared to WAN.

Disadvantages -

- ① The architecture of MAN is quite complicated hence it is hard to design & maintain.
- ② The data transfer rate in MAN is low when compare to LANs.
- ③ This network is highly expensive because it require high cost setup & fiber optics.

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WAN is a type of computer network that connects computers over large geographical distance through a shared communication task. WAN can also be defined as a group of local area network that communicate with each other. its transmission speed is low & it comes with very high maintenance & very high cost. The most common example of WAN is the internet.

Advantages -

- ① It covers large geographical area which enhances the reach of organisation to transmit data quickly.
- ② The data can be stored in centralised manner because of remote access to data provided by WAN.
- ③ WAN enables a user & or organisation to connect with world very easily & allows to exchange data & do business at global level.

Disadvantages - The data transfer rate is slow in comparison to LAN because of large distance & high number of connected system within the network.

- ① Traffic Congestion in wide area network is very high.
- ② fault tolerance ability of WAN is very less.

use of internet.

① The internet is a global networking system that can be used on most device nowadays and has become essential part of our life.

① education - most of the devices nowadays are connected through the internet. The internet has availability of broad educational content on any topic with different types.

Internet search engines help people quickly find the relevant study material in multiple format like image video^{doc}. This help eliminate the need to go to library.

② online banking → The way of banking has changed after the intodu. of the internet.

The internet has made banking online where people can manage their bank accounts while sitting at home. with the help of ^{online} banking people can securely transfer money from one account to another.

Research - Internet playing significant Role in field of Research. Before use of the internet it was quite hard to look for information about anything. people had to go through hundred of books for reference to find desired info. However the internet has made it a lot easier and anyone can find the required information.

Email- Signifi -

email has enables faster communication betⁿ people on the internet. using email people can quickly share information files images, audio, video.



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Job search - Any person can view relevant job information using internet. Many websites provides details about job availability. Once the people submit their resume that website keep information about vacancies through email.